

1. Administrative & Study Identification

**Official Study Title:** A PHASE 3, RANDOMIZED, DOUBLE BLIND, PLACEBO CONTROLLED STUDY OF PF-06821497 (MEVROMETOSTAT) WITH ENZALUTAMIDE IN METASTATIC CASTRATION RESISTANT PROSTATE CANCER (MEVPRO-2) **Plain Language Title:** A Study to Learn How PF-06821497 (Mevrometostat) Works in Men With Metastatic Castration-resistant Prostate Cancer. **Short Title/Acronym:** MEVPRO-2 **Protocol Number:** C2321003 **Posting ID:** 680142559345727201 **NCT Number:** NCT06629779 **Sponsor / Company Name:** Pfizer **Investigational Product:** PF-06821497 (Mevrometostat) + Enzalutamide **Phase of Development:** Phase III (Phase ID: PHASE3) **Trial Status:** RECRUITING (Status Description: Currently recruiting participants) **Medical Monitor:** Pfizer CT.gov Call Center (ClinicalTrials.gov\_Inquiries@pfizer.com)

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2. Study Synopsis & Rationale

**2.1 Study Objective Primary Objective:** To evaluate whether the addition of the EZH2 inhibitor mevrometostat to enzalutamide improves radiographic progression-free survival (rPFS) compared to enzalutamide plus placebo in patients with mCRPC who are androgen receptor pathway inhibitor (ARPI)-naïve. **Secondary Objectives:** To evaluate overall survival (OS), Time to Pain Progression (TTPP), Duration of Response (DoR), and the safety and tolerability of the combination therapy.

**2.2 Background & Rationale** To address the gap in extending the duration of clinical response in patients with mCRPC. The polycomb group protein enhancer of zeste homolog 2 (EZH2) is overexpressed in mCRPC and its dysregulation contributes to treatment resistance through multiple pathways, including the activation of androgen receptor transcription factors and promotion of neuroendocrine transdifferentiation. Combining an EZH2 inhibitor (mevrometostat) with an ARPI (enzalutamide) is hypothesized to delay or prevent antiandrogen resistance compared with an ARPI alone.

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3. Study Design & Methodology

**3.1 Design Framework Study Type:** Interventional **Control Type:** Placebo-controlled **Blinding:** Double-blind **Randomization:** 1:1

**3.2 Planned Enrollment & Duration Targeted Enrollment (\$N\$):** Approximately 900 male participants **Number of Sites:** ~239 locations globally (including Asia Pacific, Europe, Latin

America, North America, and South Africa) **Estimated Study Start:** 22-Oct-2024 **Estimated Primary Completion:** 02-Dec-2026

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## 4. Subject Selection (Eligibility Criteria)

**4.1 Inclusion Criteria** 1. Male patients  $\geq 18$  years of age with histologically or cytologically confirmed adenocarcinoma of the prostate without small cell features. 2. Metastatic disease in bone documented on bone scan, or in soft tissue documented on CT/MRI scan. 3. Progressive disease in the setting of medical or surgical castration (mCRPC). 4. ECOG performance status 0 or 1, with a life expectancy of  $\geq 12$  months as assessed by the investigator.

**4.2 Exclusion Criteria** 1. Any medical or psychiatric condition including recent (within the past year) or active suicidal ideation/behavior or laboratory abnormality that make the participant inappropriate for the study. 2. Prior treatment with any ARSi (enzalutamide, darolutamide, apalutamide, or abiraterone acetate) in any setting. 3. Treatment with cytotoxic chemotherapy (except docetaxel in mCRPC with no signs of failure/progression within 3 months of completion), radio-ligand therapy (i.e.  $^{177}\text{Lu}$ -PSMA-617), CDK4/6 inhibitors, 5-alpha reductase inhibitors for prostate cancer in any setting, or poly ADP-ribose polymerase (PARP) monotherapy. 4. Known history of active inflammatory gastrointestinal disease, chronic diarrhea, or previous gastric resection or lap-band surgery. 5. Known or suspected brain metastasis, active leptomeningeal disease, clinically significant history of seizure, clinically significant cardiovascular disease, or any history of myelodysplastic syndrome (MDS) or acute myeloid leukemia (AML) with a few exceptions. 6. Previous administration with an investigational product (drug or vaccine) within 30 days or 5 half-lives preceding the first dose of study intervention (whichever is longer). 7. Inadequate organ function.

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## 5. Intervention & Treatment Plan

**5.1 Investigational Regimen Dosage/Frequency:** \* **Arm A (Experimental):** Mevrometostat (875 mg twice daily with food) in combination with enzalutamide (160 mg once daily). \* **Arm B (Control):** Placebo (twice daily) in combination with enzalutamide (160 mg once daily).

*Note on Enzalutamide:* \* **Drug Preferred Name:** enzalutamide \* **Trade Name(s):** Xtandi \* **ATC Code:** L02BB04 \* **EMA URL:** [Xtandi EPAR Product Information](#)

**Route of Administration:** Oral **Duration of Treatment:** Until disease progression, unacceptable toxicity, or withdrawal of consent.

**5.2 Concomitant & Prohibited Therapy Permitted:** Ongoing androgen deprivation therapy (ADT) and first-generation anti-androgen agents. Prior docetaxel is permitted in the castrate-sensitive setting as noted above. **Prohibited:** Systemic anti-cancer treatments initiated after documentation of mCRPC (prior to study start), and any prior treatment with ARSi agents.

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## 6. Study Timeline & Visit Schedule

| Visit ID     | Window (Days) | Key Activities/Procedures                                                               |
|--------------|---------------|-----------------------------------------------------------------------------------------|
| Screening    | Day -28 to 0  | Informed Consent, Medical History, Baseline Imaging (CT/MRI/Bone Scan), Lab Assessments |
| Baseline     | Day 0         | Randomization (1:1), First Dose of Study Drug                                           |
| Interim      | Q4W (approx)  | Safety Labs, Efficacy Assessment, AE Monitoring (e.g., diarrhea, dysgeusia management)  |
| End of Study | Follow-up     | Final Vital Signs, Exit Interview, IP Accountability, Long-Term/Survival Follow-up      |

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## 7. Outcome Measures (Endpoints)

**7.1 Primary Endpoint Definition:** Radiographic Progression Free Survival (rPFS), defined as the time from the date of randomization to first objective evidence of radiographic progression as assessed in soft tissue per Response Evaluation Criteria in Solid Tumors (RECIST) v1.1 or in bone per Prostate Cancer Clinical Trials Working Group 3 (PCWG3) guidelines by BICR, or death, whichever occurs first.

**7.2 Secondary Outcomes** \* **Overall Survival (OS)** \* **Time to Pain Progression (TTPP):** To demonstrate that PF-06821497 with enzalutamide is superior to placebo with enzalutamide in prolonging TTPP. \* **Duration of Response (DoR):** Assessed in measurable soft tissue disease. \* **Safety Profile:** Safety and adverse event assessment.

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## 8. Safety & Adverse Event Reporting

**Adverse Event (AE) Monitoring:** Standard clinical oncology monitoring, with a specific focus on manageable AEs associated with the investigational combination, such as diarrhea, dysgeusia, and anemia. **Serious Adverse Events (SAEs):** Reported within standard regulatory timelines (typically 24 hours). **Data Monitoring Committee (DMC):** External independent committee reviewing blinded safety data globally.

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## 9. Statistical Analysis Plan (SAP) Summary

**Analysis Populations:** Intent-to-Treat (ITT) population for primary efficacy endpoints (rPFS, OS). Safety Set for all participants receiving at least one dose of the study drug.

**Sample Size Justification:** Target  $N = 900$  randomized 1:1, adequately powered to detect a statistically significant improvement in rPFS for the combination arm. **Handling of Missing Data:** Standard oncology methods for time-to-event endpoints; right-censoring at the last evaluable tumor assessment for dropouts prior to documented progression.

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## 10. Quality Assurance & Regulatory Compliance

**GCP Compliance:** This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.

**Monitoring Plan:** Standard risk-based onsite and remote monitoring schedules per sponsor guidelines. **Audit Trail:** Validated Electronic Data Capture (EDC) system with eCRF locking, query management, and centralized imaging reviews (BICR).

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## 11. Study Identifiers & Governance

**Primary ID:** C2321003 **Registry Number:** NCT06629779 **Sponsor/Lead Organization:** Pfizer **Criteria Type & Text:** PHASE\_REQUIREMENT (Trial must be in PHASE3) **Source Level:** 5

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## 12. Clinical Framework (mCRPC Specific)

**Disease Area / Preferred UMLS Name:** Metastatic Castration-Resistant Prostate Cancer (mCRPC) **Semantic Type:** Neoplastic Process **Definition:** A prostate carcinoma that has spread to other sites and is resistant to castration therapy. **MeSH Code:** D000085434 **SNOMED ID:** 789498007 **SNOMED Hierarchy:** 126909001 | Castration-resistant prostate cancer | Castration-resistant prostate cancer (disorder) **Medical Methodology:** Androgen Receptor Pathway Inhibitor (ARPI) combined with Epigenetic Modulation (EZH2 inhibition) **Optimization Target:** Delay or prevention of antiandrogen resistance and extension of duration of clinical response.

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## 13. Study Procedures & Methodology

**Management Pathway:** Standard oncology pathway for mCRPC with standardized imaging. **Education Protocol (HCP):** AE management protocols (focusing on GI toxicities like diarrhea) and dose modification guidelines. **Education Protocol (Patient):** Oral oncolytic adherence and proactive symptom reporting. **Quality**

**Audit Mechanism:** Blinded Independent Central Review (BICR) of all tumor imaging.

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**14. Observation & Follow-Up Schedule**

**Total Duration:** Until estimated study completion (November 2028)  
**Onsite Visit Cadence:** Standard treatment cycles. **Remote Monitoring Frequency:** Based on central clinical operations directives. **Checklist Review Interval:** Regular imaging reviews (CT/MRI/Bone Scans) per PCWG3 and RECIST 1.1 scheduling protocols.

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**15. Sign-off & Approval**

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|--------------------------|--------|-----------|---------------|-----|-----|------|-----|
| Role                     | Name   | Signature | Date          | :-- | :-- | :--  | :-- |
| Principal Investigator   | [Name] | _____     | [DD-MMM-YYYY] |     |     |      |     |
| Clinical Operations Lead | [Name] | _____     | [DD-MMM-YYYY] |     |     | Lead |     |
| Statistician             | [Name] | _____     | [DD-MMM-YYYY] |     |     |      |     |